

1. Evaluate the following:

a. -2^{-3}

b. $\left(-\frac{2}{3}\right)^{-2}$

c. $8^{-4/3}$

d. $f(x) = 5^{2-3x}$, find $f(1)$.

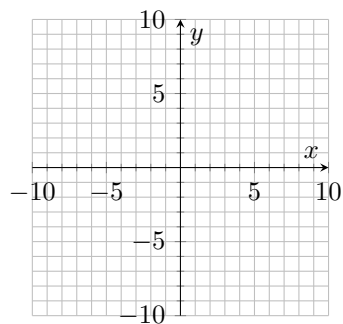
e. $\lim_{x \rightarrow \infty} 3^x$

f. $\lim_{x \rightarrow -\infty} 3^x$

g. $\lim_{x \rightarrow \infty} 3^{-x}$

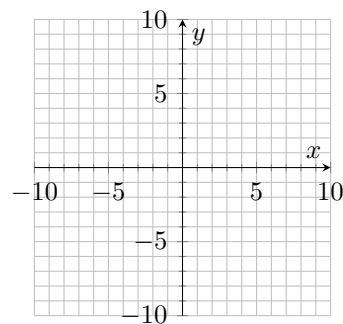
2. Graph the following:

a. $f(x) = 3^{-x}$



i. What is the domain and range of $f(x)$?

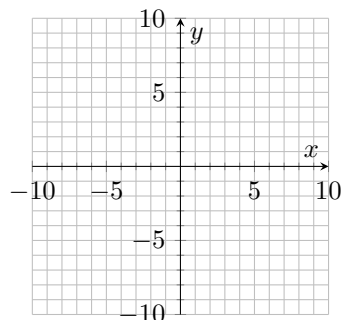
b. $g(x) = 3^{-(x-1)} + 2$



i. What is the domain and range of $g(x)$?

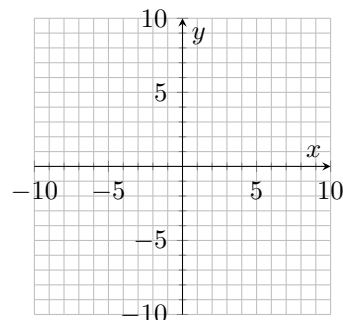
3. Graph the following:

a. $f(x) = \log_3 x$



i. What is the domain and range of $f(x)$?

b. $g(x) = \log_3 (x - 1)$



i. What is the domain and range of $g(x)$?

4. Evaluate the following:

a. $\left(\frac{1}{6}\right)^x = 36$

b. $\log_3(27)$

c. $\log_{\sqrt{10}} 1$

d. $\ln\left(\frac{1}{e^2}\right)$

e. Write $z = \log_6 5$ as an exponential equation.

f. $\lim_{x \rightarrow 0^+} \log_3 x$

5. If \$1000 is invested in an account at 5% annual interest compounded monthly, what is the amount in the account after 6 years?

6. Solve for x . Write your answer as an exact value and approximate value.

a. $5^x = 2$

b. $(3.5)^{2x} = 10$

c. $2 \ln x = 7$

d. $\log_2 x = \sqrt{5}$

e. $(1 + x)^4 = 3.1$ (only approximate)

7. If \$1000 is invested in an account at 5% compounded continuously, how long will it take to double the investment?

8. Use the properties of logarithms to expand the expression as much as possible.

f. $\log_2(8x)$

g. $\ln\left(\frac{e^6}{11}\right)$

h. $\ln\left(\frac{\sqrt[3]{xy}}{t^2}\right)$

9. Use the properties of logarithms to rewrite the expression as a single logarithm.

i. $\log_2(5) + 3\log_2(x)$

j. $\frac{1}{3}\ln(6) - \frac{1}{3}\ln(2)$

k. $\log(2) + \log(3) + \log(y) - \log(z)$

l. $\frac{1}{2}[\log(x) + \log(y)] - \log(z)$